

CSE331: Automata & Computability

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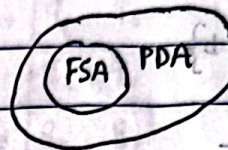
Theme: auto nom nu20 1a20

TOC → Theory of Computation

- Automata
- Computability Theorem
- Complexity Theorem

Automata:

- Conceptual / Theoretical Machine
- FSA - Finite State Automation
- PDA - Push Down Automation



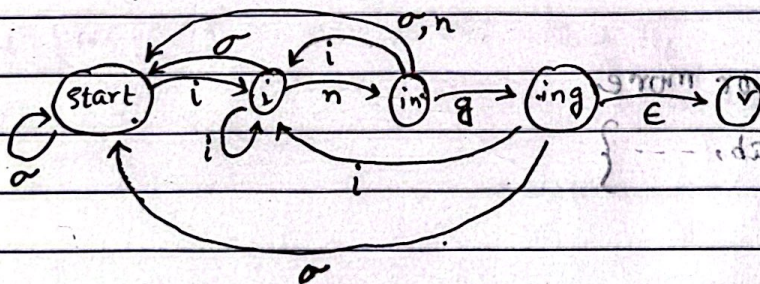
* Finite State Automata

- Deterministic
- Non-deterministic

Symbols

Ex: Ends with "ing"

Given string = "lat initiationg"



let's run "innings"

Symbols

a, b, c, --

0, 1, 2 --

strings / Sentence

ε, 10100110, abc, cab, bbacc

Alphabets

$\Sigma = \{0, 1\}$

$= \{a, b, c\}$

$= \{0, 1, 2, \dots\}$

Language = $\{01, 101, 1101, 1111001, abc\}$

⊙ Symbols → Alphabets → strings → Language

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Concatenation:

$$ab \cdot cd \rightarrow abcd$$

$$L(A) = \{0, 1\} \quad L(B) = \{a, b\}$$

$$L(C) = L(A) \cdot L(B)$$

$$= \{0a, 0b, 1a, 1b\}$$

$$\Sigma = \{0, 1, a, b\}$$

Kleen Closure

$$\Sigma^* \rightarrow 0 \text{ or more}$$

$$\Sigma' = \{a, b\}$$

$$L^0 = \{\epsilon\}$$

$$L^1 = \{a, b\}$$

$$L^2 = \{aa, ab, ba, bb\}$$

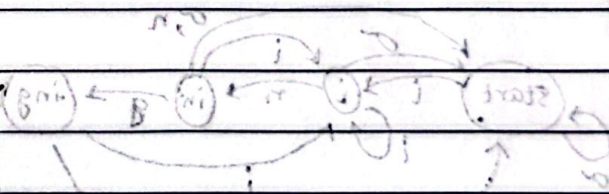
$$L^0 \cup L^1 \cup L^2 \cup \dots$$

$$= \{\epsilon, a, b, aa, bb, ab, ba, aaa, \dots\}$$

Pos Positive Closure

$$\Sigma^+ \rightarrow 1 \text{ or more}$$

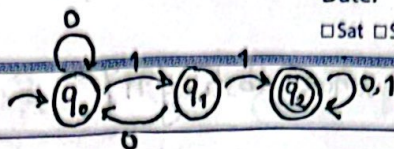
$$\rightarrow \{a, b, aa, ab, \dots\}$$



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$M = (Q, \Sigma, \delta, q_0, F)$



$L_M = \{\text{strings containing 11}\}$

Q is the set of states.

$Q = \{q_0, q_1, q_2\}$

Σ is the alphabet.

$\Sigma = \{1, 0\}$

δ is the transition function.

$\delta:$

q_0 is the start state.

q_0

F is the set of accept/final state.

q_1

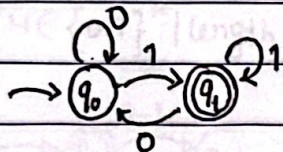
q_2

$F = \{q_2\}$

Examples:

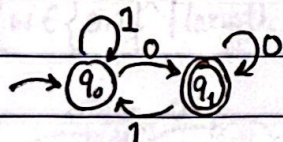
1. $L_1 = \{w \in \{0, 1\}^* \mid w \text{ ends with a 1}\}$

$\Rightarrow M_1:$



2. $L_2 = \{w \in \{0, 1\}^* \mid w \text{ does not end with a 1}\}$

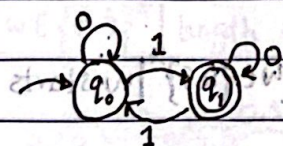
$\Rightarrow M_2:$



$\{0, 10, 0110, 101100, \dots\}$

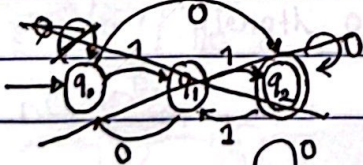
3. $L_3 = \{w \in \{0, 1\}^* \mid w \text{ has odd number of 1's}\}$

$\Rightarrow M_3:$

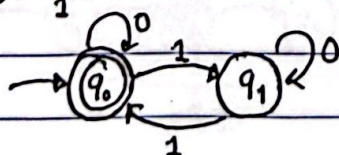


4. $L_4 = \{w \in \{0, 1\}^* \mid w \text{ has even number of 1's}\}$

$\Rightarrow M_4:$



$0, 011, 101, 110$

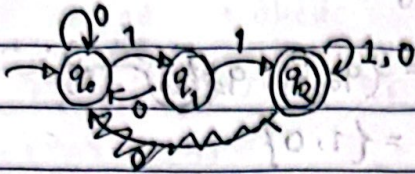


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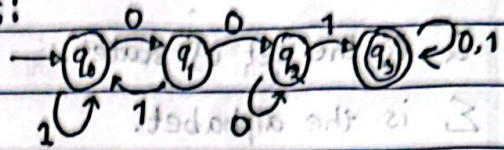
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5. $L_5 = \{w \in \{0,1\}^* \mid w \text{ has substring } 11\}$ 6. $L_6 = \{w \in \{0,1\}^* \mid w \text{ has substring } 001\}$

$\Rightarrow M_5:$

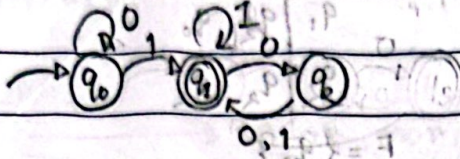


$\Rightarrow M_6:$



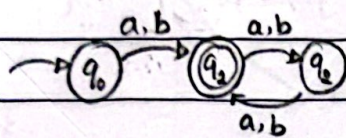
7. $L_7 = \{w \in \{0,1\}^* \mid \text{contain at least one 1 and even number of 0's follow the last 1}\}$

$\Rightarrow M_7:$

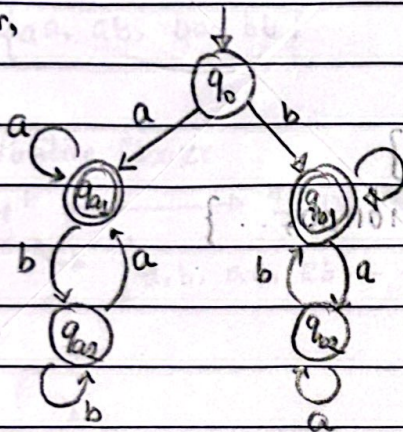


8. $L_8 = \{w \in \{a,b\}^* \mid w \text{ starts and ends with a or } w \text{ starts and ends with b}\}$

$\Rightarrow M_8:$

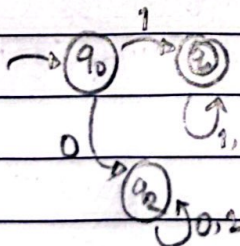


Or,



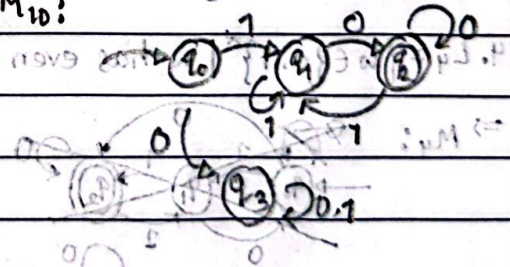
9. $L_9 = \{w \in \{0,1\}^* \mid w \text{ starts with 1}\}$

$\Rightarrow M_9:$



10. $L_{10} = \{w \in \{0,1\}^* \mid w \text{ starts with 1 and ends with 0}\}$

$\Rightarrow M_{10}:$

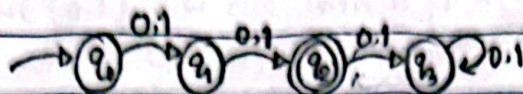


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11. $L_{11} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is two}\}$

$\Rightarrow M_{11}$:



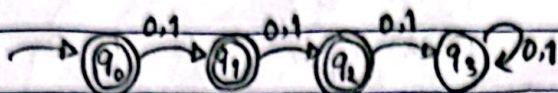
12. $L_{12} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is at least two}\}$

$\Rightarrow M_{12}$:



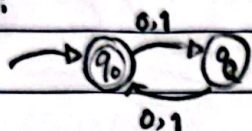
13. $L_{13} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is at most two}\}$

$\Rightarrow M_{13}$:



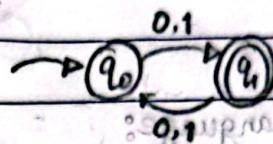
14. $L_{14} = \{w \in \{0,1\}^* \mid w \text{ has even length}\}$

$\Rightarrow M_{14}$:



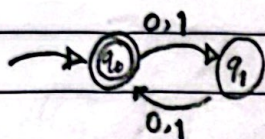
15. $L_{15} = \{w \in \{0,1\}^* \mid w \text{ has odd length}\}$

$\Rightarrow M_{15}$:



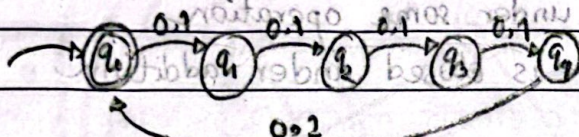
16. $L_{16} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is multiple of 2}\}$

$\Rightarrow M_{16}$:



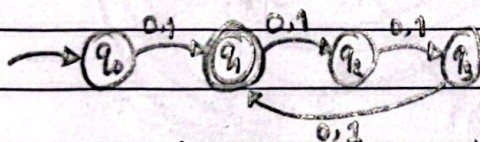
17. $L_{17} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is multiple of 5}\}$

$\Rightarrow M_{17}$:



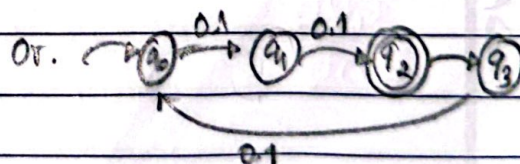
18. $L_{18} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is } 3n+2 \text{ where } n \geq 0\}$

$\Rightarrow M_{18}$:



19. $L_{19} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is 2 less than multiple of 4}\}$

$\Rightarrow M_{19}$:



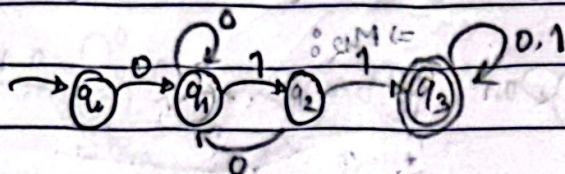
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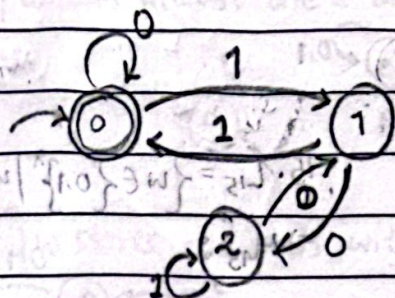
* $L = \{w \in \{0,1\}^* \mid w \text{ contains } 011\}$

$\Rightarrow M:$



* $L = \{w \in \{0,1\}^* \mid w, \text{ when interpreted as a binary number, is divisible by } 3\} \cup \{\epsilon\}$

$\Rightarrow M:$



Regular Language:

- Union

- Concatenation

- Kleene star

Regular languages are closed under Union

"closed under some operation"

The set of integers is closed under addition

* L_1 Regular, L_2 Regular $\Rightarrow L_1 \cup L_2$ Regular

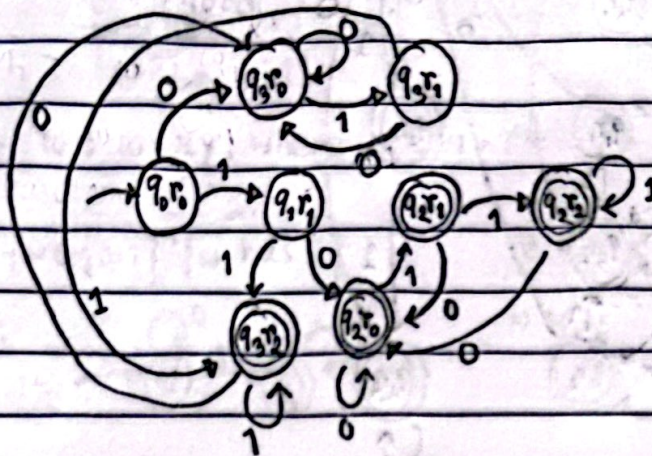
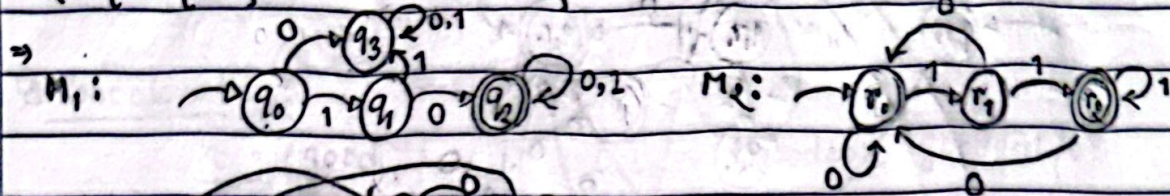
L_1 "starts with 10" $10(01)^*$

L_2 "ends with 11" $(01)^*11$

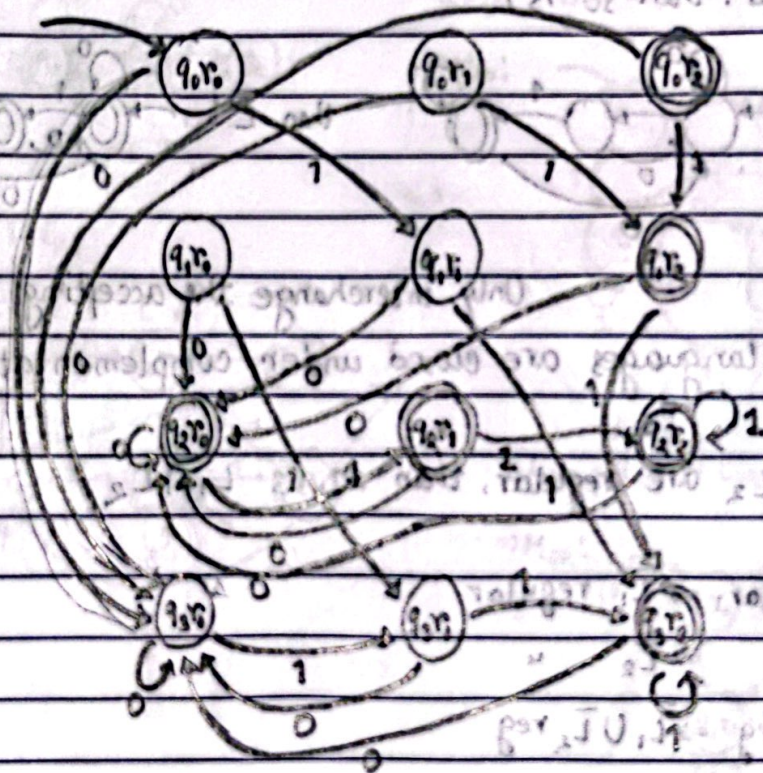
$L_1 \cup L_2$ $10(01)^* \cup (01)^*11$

$$L_1 = \{w \in \{0,1\}^* \mid w \text{ starts with } 10\}$$

$$L_2 = \{w \in \{0,1\}^* \mid w \text{ ends with } 11\}$$



1011000011



The language accepted by $M_1 \times M_2$ is the intersection of L_1 and L_2 , which is the set of strings that start with 10 and end with 11.

The language accepted by $M_1 \times M_2$ is the intersection of L_1 and L_2 .

Using Product Construction

$\therefore L_1 \cup L_2$ is regular

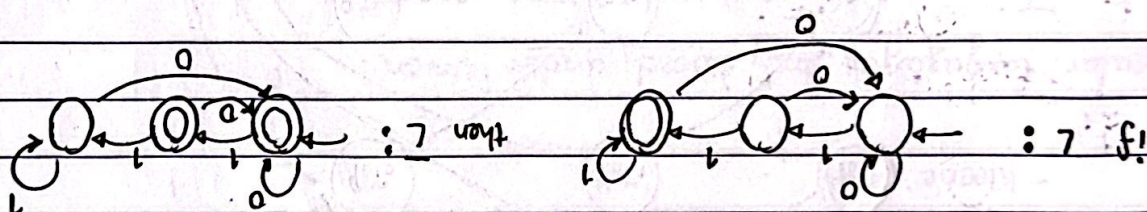
$\overline{L_1 \cup L_2} = \overline{L_1} \cap \overline{L_2}$ (De Morgan's Law)

$L_1 \text{ reg}, L_2 \text{ reg} \Rightarrow \overline{L_1} \cup \overline{L_2} \text{ reg}$

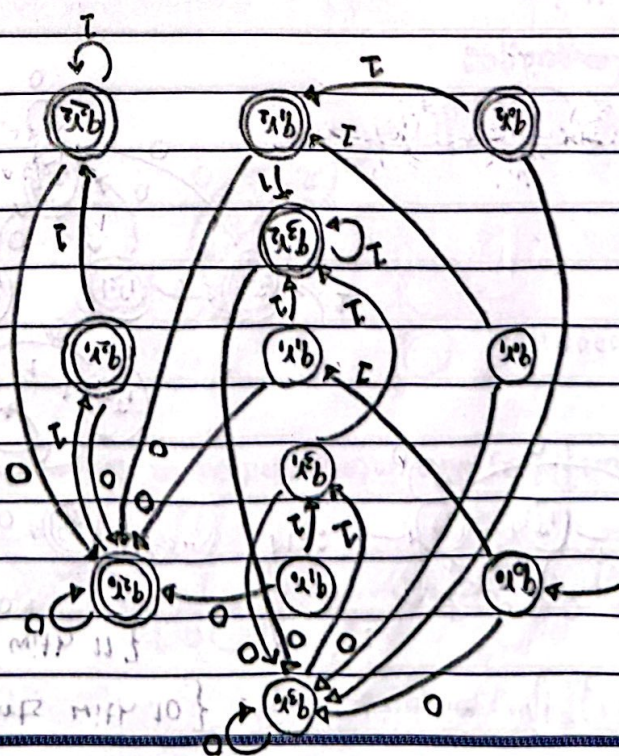
$L_2 \cup \dots \cup L_n \Rightarrow L_2 \text{ reg}$
 $L_1 \text{ is regular} \Rightarrow \overline{L_1} \text{ regular}$

* If L_1 and L_2 are regular, then so is $L_1 \cup L_2$

* Regular languages are closed under complementation.
 Only interchange the accepting state.



* If L is regular, then so is $\overline{L}$



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⊛ If L_1 and L_2 is regular then so is $L_1 \setminus L_2$

$L_1 \cap \bar{L}_2$

⊛ Concatenation:

$L_1 = \{\text{good}\}$

$L_1 L_2 = \{\text{goodboy, goodgirl}\}$

$L_2 = \{\text{boy, girl}\}$

$L_1 L_2 = \{w : w = xy, \text{ where } x \in L_1, y \in L_2\}$

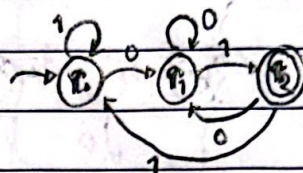
⊛ $L_{17} = \{w \in \{0,1\}^* \mid w \text{ has } 1\}$

⊛ $L_{18} = \{w \in \{0,1\}^* \mid w \text{ end with } 01\}$

$\Rightarrow M_{17}:$

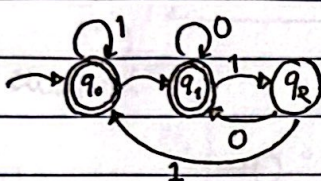


$\Rightarrow M_{18}:$



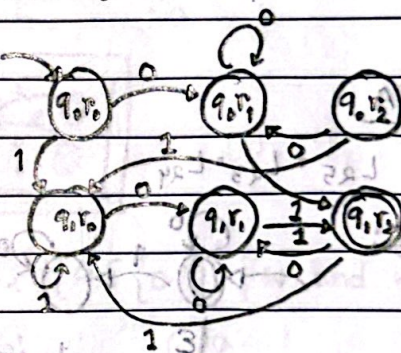
⊛ $L_{19} = \bar{L}_{18}$

$\Rightarrow M_{19}:$



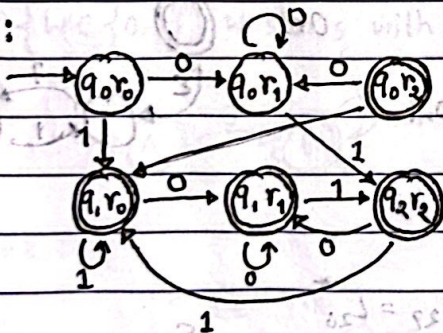
⊛ $L_{20} = L_{17} \cap L_{18}$

$\Rightarrow M_{20}:$



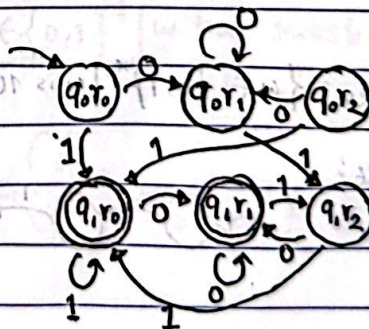
⊛ $L_{21} = L_{17} \cup L_{18}$

$\Rightarrow M_{21}:$



⊛ $L_{22} = L_{17} \oplus L_{18}$

$\Rightarrow M_{22}:$



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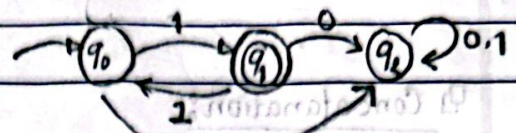
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$L_{23} = \{w \in \{0,1\}^* \mid 0^n \text{ where } n \geq 0\}$

$\Rightarrow M_{23}$:

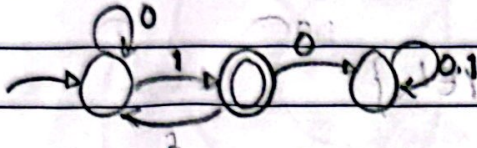


$\Rightarrow M_{24}$:



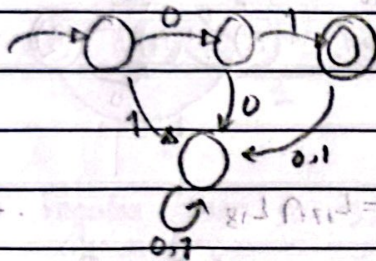
$L_{25} = L_{23} \circ L_{24}$

$\Rightarrow M_{25}$:



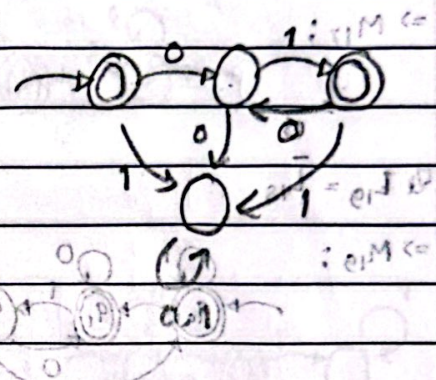
$L_{26} = \{w \in \{0,1\}^* \mid w \text{ is } 10\}$

$\Rightarrow M_{26}$:



$L_{27} = L_{26}^*$

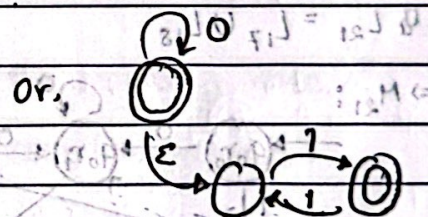
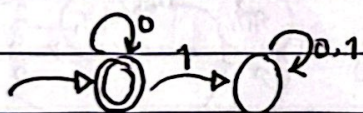
$\Rightarrow M_{27}$:



NFA

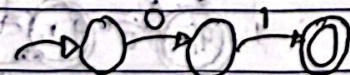
For $L_{25} = L_{23} \circ L_{24}$

$\Rightarrow M_{25}$:



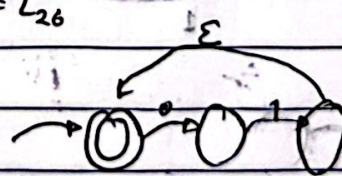
$L_{26} = \{w \in \{0,1\}^* \mid w \text{ is } 10\}$

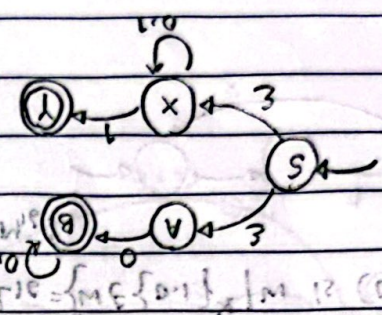
$\Rightarrow M_{26}$:



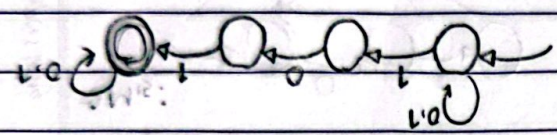
$L_{27} = L_{26}$

$\Rightarrow M_{27}$:

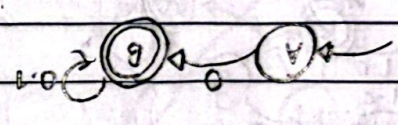




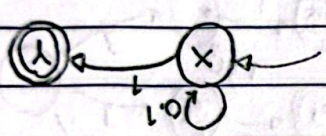
$L_3 = L_1 \cup L_2$
 $M_3 = \{M_1 \cup M_2\}$
 $M_3 = \{M_1 \cup M_2\}$



$L_4 = \{w \in \{0,1\}^* \mid w \text{ has substring } 101\}$
 $M_4 = \{M_1 \cup M_2\}$
 $M_4 = \{M_1 \cup M_2\}$

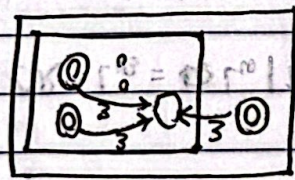


$M_1 = \{M_1\}$

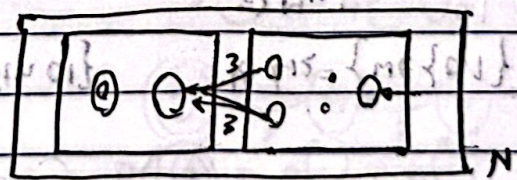


$M_2 = \{M_2\}$

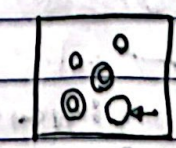
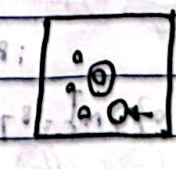
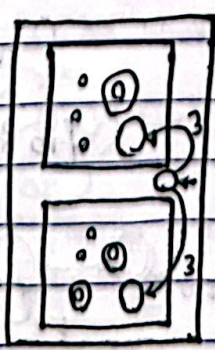
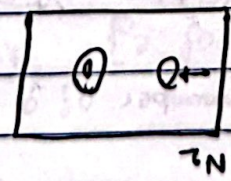
$L_1 = \{w \in \{0,1\}^* \mid w \text{ starts with } 0\}$
 $L_2 = \{w \in \{0,1\}^* \mid w \text{ ends with } 1\}$



• Kleene closure (*)



• Concatenation (o)



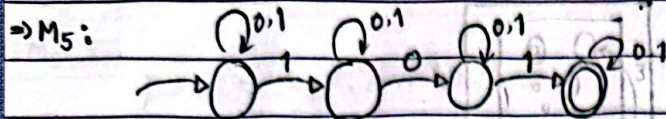
• OR/Union ($\cup$)

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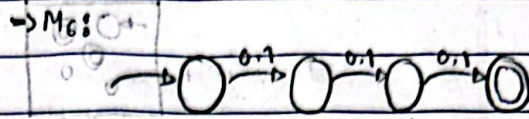
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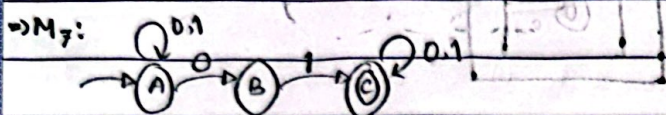
$L_5 = \{w \in \{0,1\}^* \mid w \text{ has subsequence } 101\}$



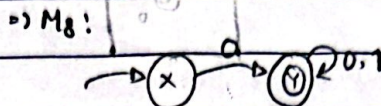
$L_6 = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is } 3\}$



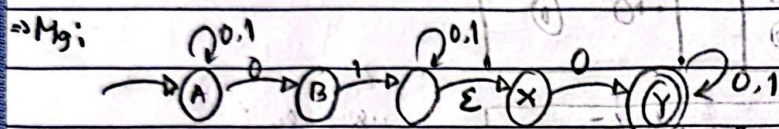
$L_7 = \{w \in \{0,1\}^* \mid w \text{ has substring } 01\}$



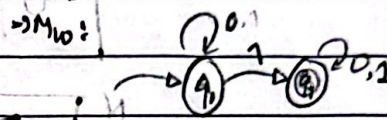
$L_8 = \{w \in \{0,1\}^* \mid w \text{ starts with } 0\}$



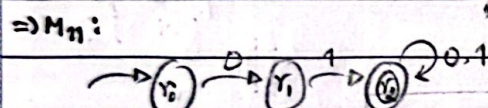
$L_9 = L_7 \cup L_8$



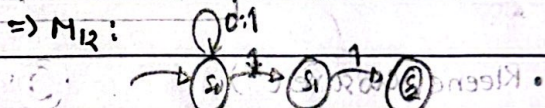
$L_{10} = \{w \in \{0,1\}^* \mid w \text{ has } 1\}$



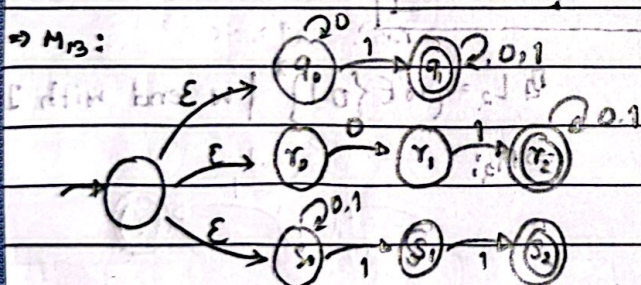
$L_{11} = \{w \in \{0,1\}^* \mid w \text{ starts with } 01\}$



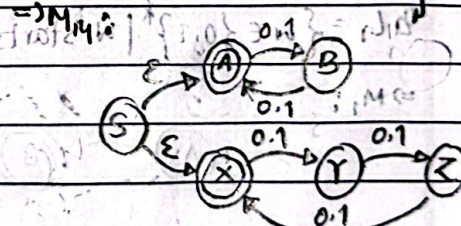
$L_{12} = \{w \in \{0,1\}^* \mid w \text{ ends with } 11\}$



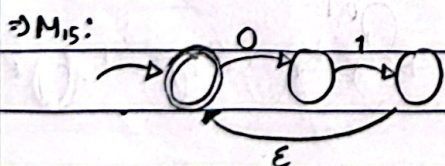
$L_{13} = L_{10} \cup L_{11} \cup L_{12}$ OR $L_{13} = L_{10} \cup L_{11} \cup L_{12}$



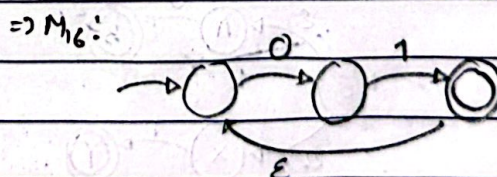
$L_{14} = \{w \in \{0,1\}^* \mid \text{length of } w \text{ is multiple of } 2 \text{ or } 3\}$



$L_{15} = \{w \in \{0,1\}^* \mid w \text{ is } (01)^m, m \geq 0\}$

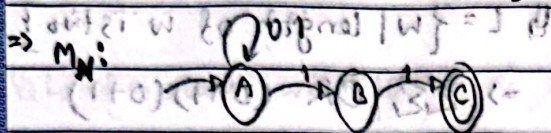


$L_{16} = \{w \in \{0,1\}^* \mid w \text{ is } (01)^m, m \geq 1\}$

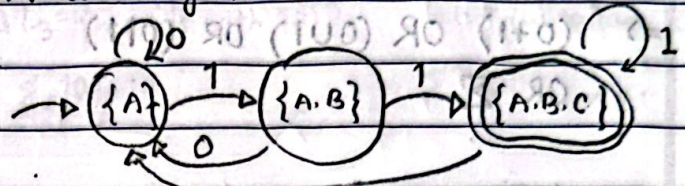


NFA to DFA

$L = \{w \in \{0,1\}^* \mid w \text{ end with } 11\}$



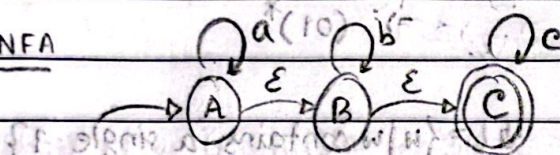
Now converting to DFA



ϵ -NFA to NFA to DFA

$L = \{w \in \{a,b,c\}^* \mid w \text{ has some number of } a \text{ followed by some number of } b \text{ followed by some number of } c\}$

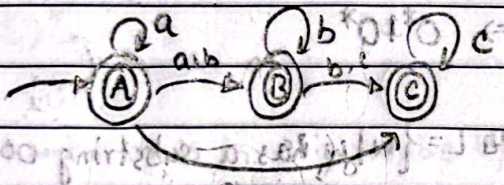
$\Rightarrow$ ϵ -NFA



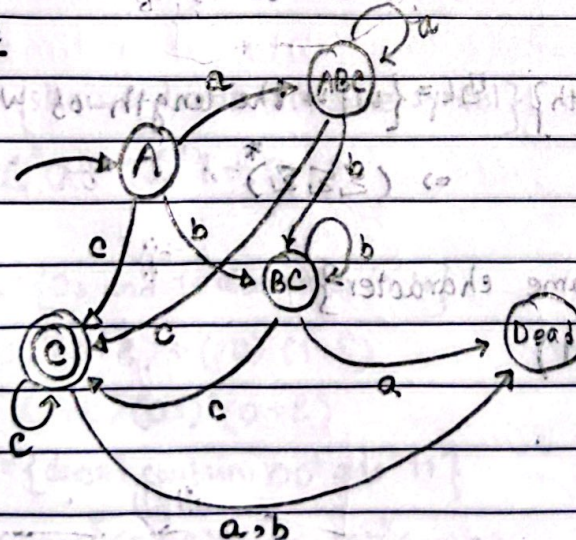
Transition table

states	a	b	c
A	{A,B,C}	{B,C}	{C}
B	{}	{B,C}	{C}
C	{}	{}	{C}

NFA



DFA



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⇒ Regular Expression:

Let, $\Sigma = \{0, 1\}$

$L = \{w \mid \text{length of } w \text{ is one}\}$

⇒ $(0+1)$ OR $(0+1)$ OR $(0+1)$
OR Σ

$L = \{w \mid \text{length of } w \text{ is two}\}$

⇒ $\Sigma\Sigma$ OR $(0+1)(0+1)$

$L = \{w \mid \text{length of } w \text{ is 0 or more}\}$

⇒ $(0+1)^*$

$L = \{w \mid \text{length of } w \text{ is 1 or more}\}$

⇒ $(0+1)^+$

$L = \{w \mid w \text{ is any number of 0's}\}$

⇒ 0^*

$L = \{w \mid w \text{ is any number of 01 substrings}\}$

⇒ $(01)^*$

$L = \{w \mid w \text{ starts with 0}\}$

⇒ $0(0+1)^*$

$L = \{w \mid w \text{ contains a single 1}\}$

⇒ 0^*10^*

$L = \{w \mid w \text{ has at least one 1}\}$

⇒ $(0+1)^*1(0+1)^*$

$L = \{w \mid w \text{ has a substring 001}\}$

⇒ $(0+1)^*001(0+1)^*$

$L = \{w \mid w \text{ is a string of even length}\}$

⇒ $((0+1)(0+1))^*$ OR $(\Sigma\Sigma)^*$

$L = \{w \mid \text{the length of } w \text{ is multiple of 3}\}$

⇒ $(\Sigma\Sigma\Sigma)^*$

$L = \{w \mid w \text{ starts and end with same character}\}$

⇒ $(0(0+1)^*0 + 1(0+1)^*1 + 0 + 1)$

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$L_0 = \{w \mid w \text{ contains } 101 \text{ as a substring}\}$ $L_1 = \{w \mid w \text{ contains } 101 \text{ as a subsequence}\}$

$$\Rightarrow (0+1)^* 101 (0+1)^*$$

$$\Rightarrow \Sigma^* 1 \Sigma^* 0 \Sigma^* 1 \Sigma^*$$

$L_2 = \{w \text{ starts with } 101\}$

$$\Rightarrow 101 \Sigma^*$$

$L_3 = \{w \text{ ends with } 101\}$

$$\Rightarrow \Sigma^* 101$$

$L_4 = \{w \text{ contains } 00 \text{ or } 11\}$

$$\Rightarrow \Sigma^* (00+11) \Sigma^*$$

$L_5 = \{w \text{ contains exactly two } 1\text{'s}\}$

$$\Rightarrow 0^* 10^* 10^*$$

$L_6 = \{w \text{ contains at least two } 1\text{'s}\}$

$$\Rightarrow \Sigma^* 1 \Sigma^* 1 \Sigma^*$$

$L_7 = \{\text{at most two } 1\text{'s}\}$

$$\Rightarrow 0^* (1+\epsilon) 0^* (1+\epsilon) 0^*$$

$L_8 = \{\text{length of } w \text{ is even or divisible by } 2\}$

$$\Rightarrow (\Sigma \Sigma)^*$$

$L_9 = \{\text{length of } w \text{ is odd}\}$

$$\Rightarrow \Sigma (\Sigma \Sigma)^*$$

$L_{10} = \{\text{length of } w \text{ is divisible by } 3\}$

$$\Rightarrow (\Sigma \Sigma \Sigma)^*$$

$L_{11} = \{\text{length of } w \text{ is not divisible by } 3\}$

$$\Rightarrow \Sigma (\Sigma \Sigma \Sigma)^* + \Sigma \Sigma (\Sigma \Sigma \Sigma)^*$$

$L_{12} = \{\text{number of } 1\text{'s in } w \text{ is divisible by } 3\}$

$$\Rightarrow (0^* 10^* 10^* 10^*)^*$$

$L_{13} = \{\text{starts and ends with different symbol}\}$

$$\Rightarrow 0 \Sigma^* 1 + 1 \Sigma^* 0$$

$L_{14} = \{\text{starts and ends with same symbol}\}$

$$\Rightarrow 0 \Sigma^* 0 + 1 \Sigma^* 1 + 0 + 1$$

$L_{15} = \{w \text{ doesn't end with } 01\}$

$$\Rightarrow \Sigma^* (00+10+11) + 0+1+\epsilon$$

$L_{16} = \{0\text{'s and } 1\text{'s alternate in } w\}$

$$\Rightarrow (01)^* (0+\epsilon) + (10)^* (1+\epsilon)$$

$L_{17} = \{\text{doesn't contain } 00\}$

$$\Rightarrow (1+01)^* (0+\epsilon)$$

$L_{18} = \{\text{doesn't contain } 10\}$

$$\Rightarrow 0^* 1^*$$

$$\text{OR, } (1+\epsilon) (01)^* (0+\epsilon)$$

$L_{19} = \{\text{doesn't contain } 00 \text{ and } 11\}$

$$\Rightarrow (1+01)^* (0+\epsilon) \cap (0+10)^* (1+\epsilon)$$

$L_{20} = \{\text{contains } 0 \text{ in every third position}\}$

$$\Rightarrow (\Sigma \Sigma 0)^* (\epsilon + \Sigma + \Sigma \Sigma)$$

$$\text{OR } (1+\epsilon) (01)^* (\epsilon + 0)$$

$$\text{OR } (01)^* (0+\epsilon) + (10)^* (1+\epsilon)$$

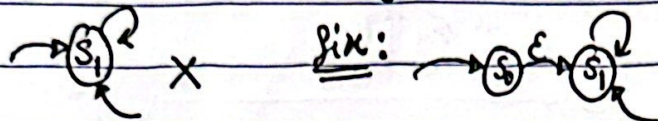
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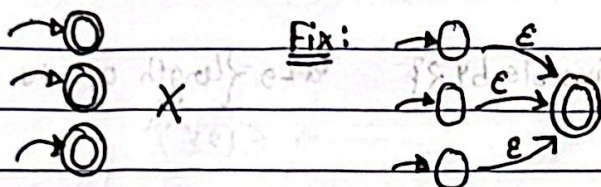
⇒ DFA to Regular Expression State Elimination Method :

* Pre-Processing :

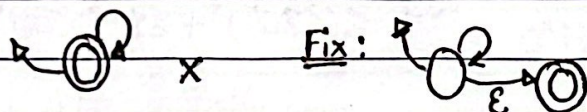
- Step 1: Can't have incoming arrows in the start state



- Step 2: Can't have multiple end state.

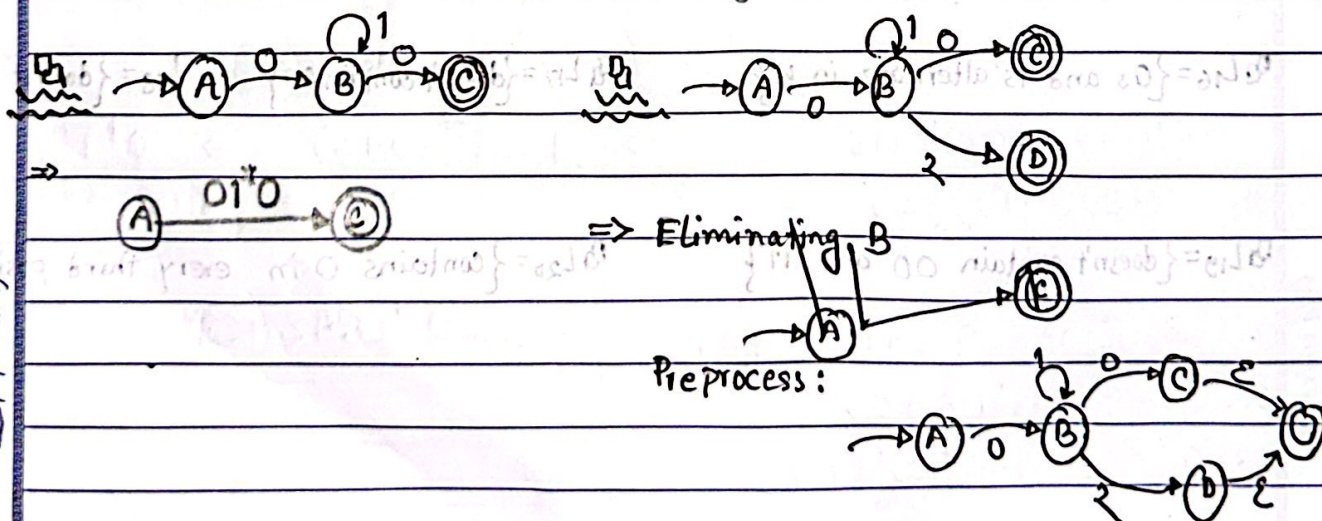


- Step 3: Can't have outgoing arrows in the accepting state.



* Main Process: While there is a state between the start and end state we will eliminate one state at a time and do some damage control.

Regular Expression Construction

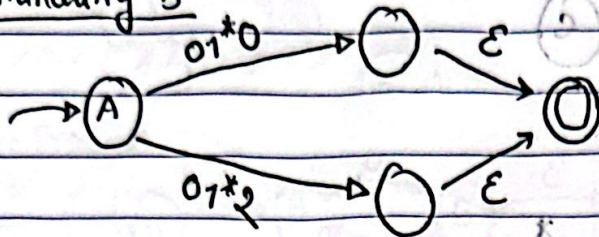


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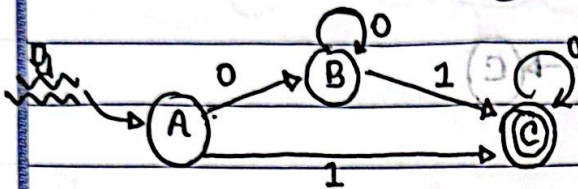
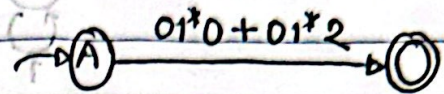
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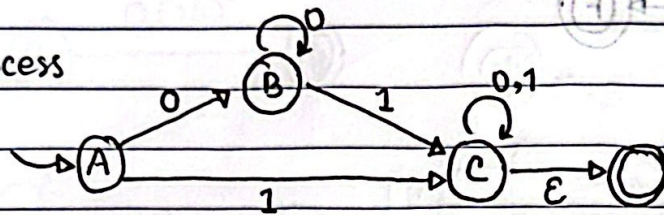
Eliminating B



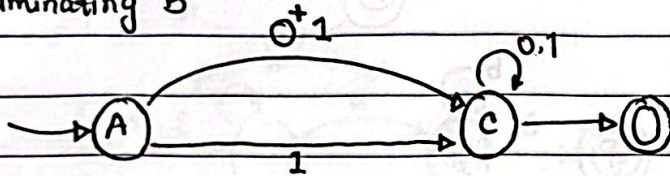
Eliminating C & D



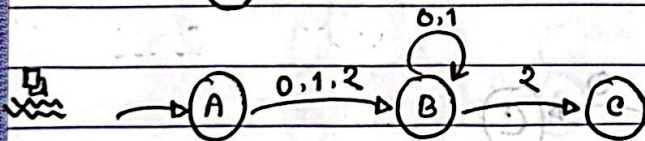
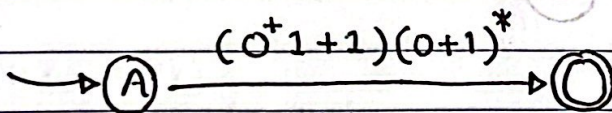
$\Rightarrow$ Pre Process



Eliminating B

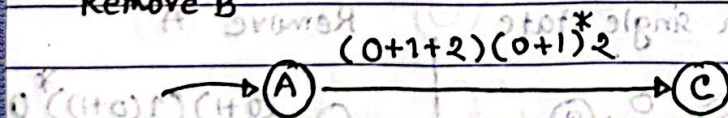


Eliminating C



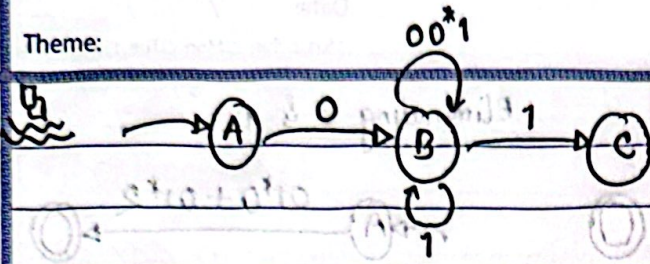
$\Rightarrow$

Remove B

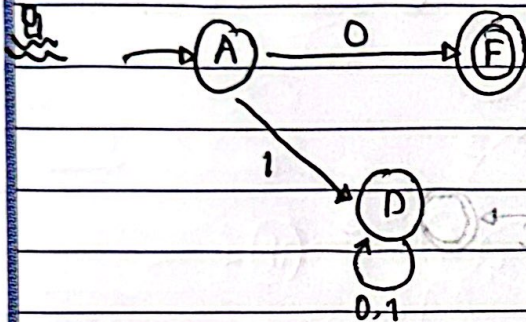
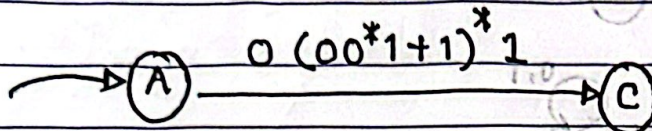


Theme:

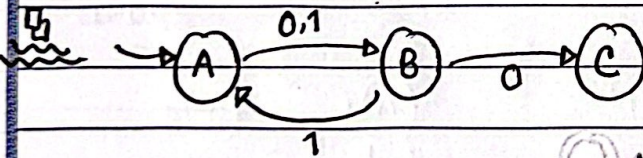
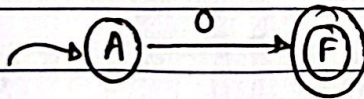
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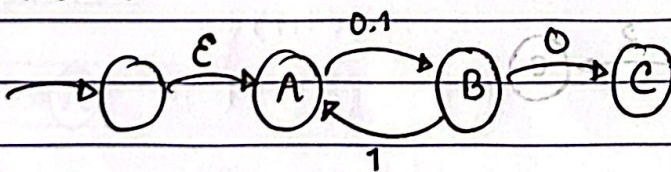
⇒ Remove B



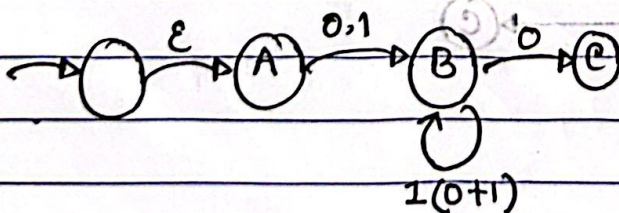
⇒ Remove D.



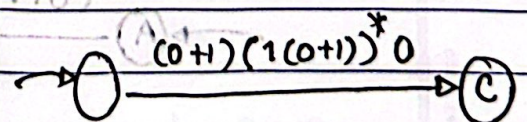
⇒ Pre Process



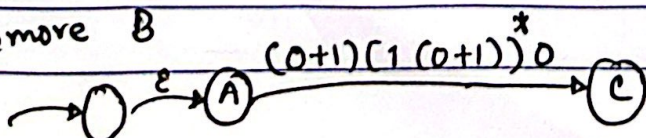
Replacing multistate loop to a single state



Remove A



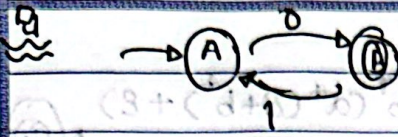
Remove B



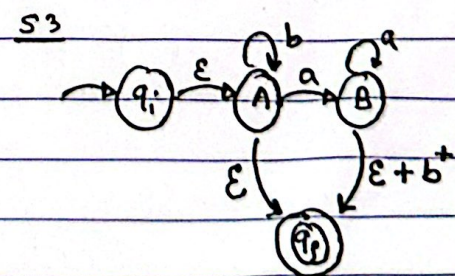
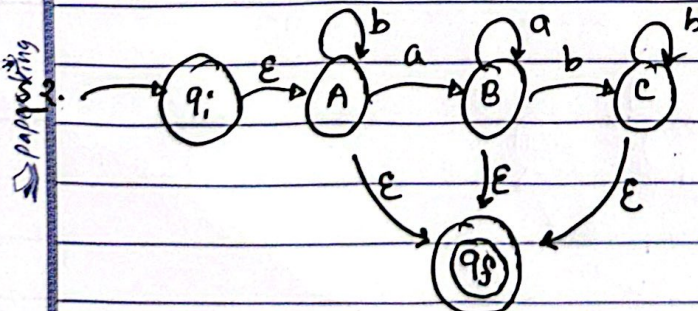
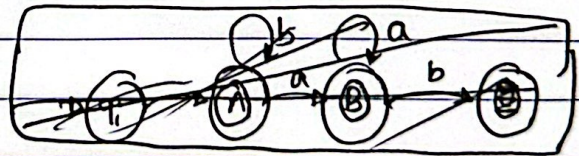
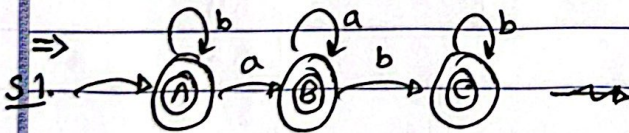
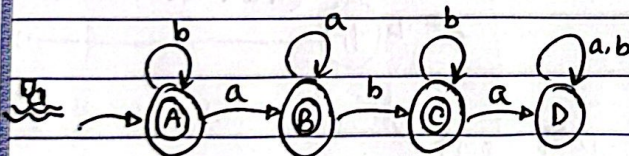
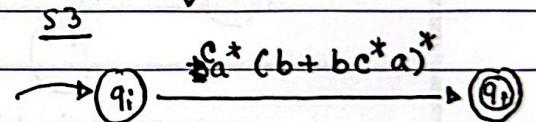
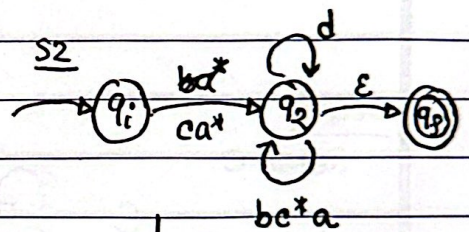
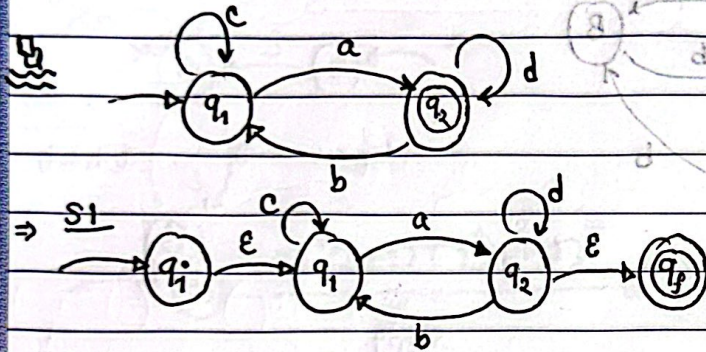
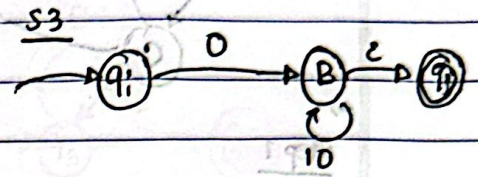
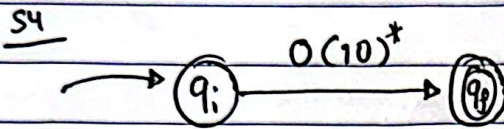
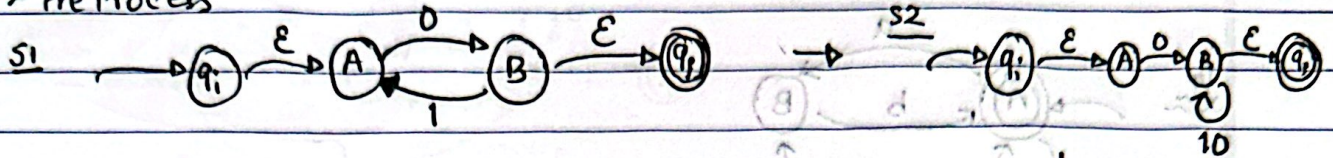
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→ Pre Process

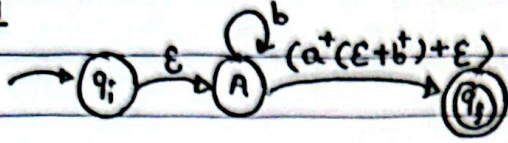


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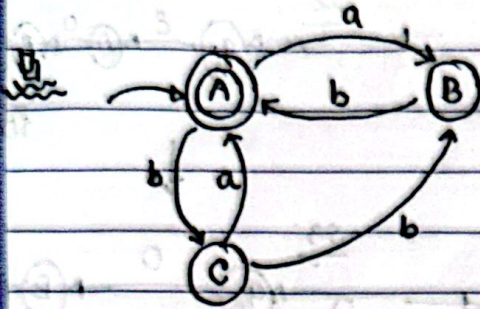
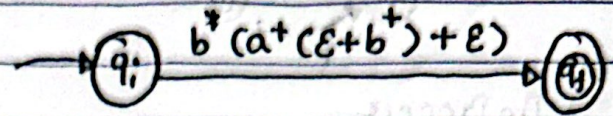
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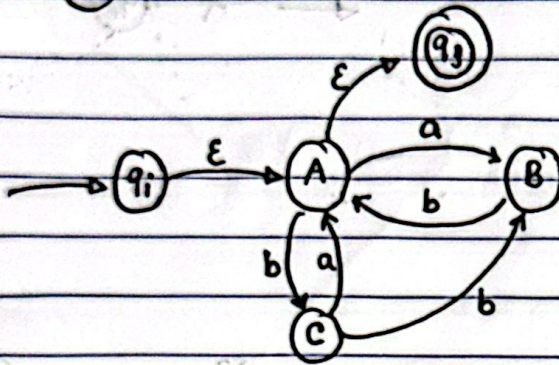
S4



S5

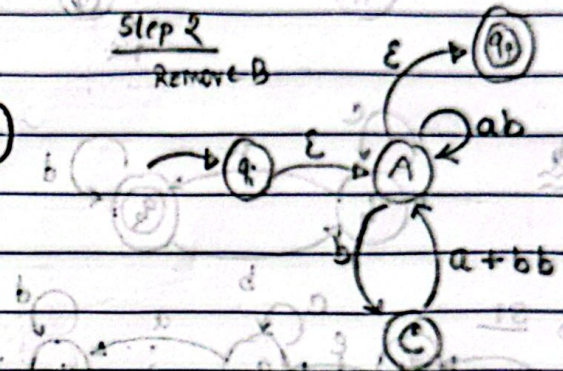


Step 1



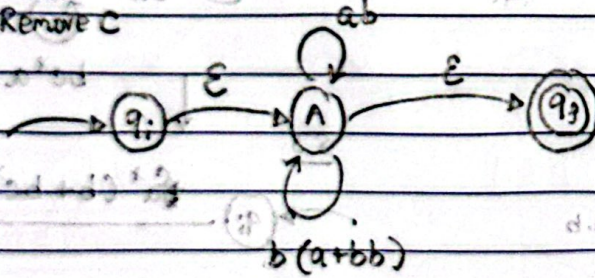
Step 2

Remove B



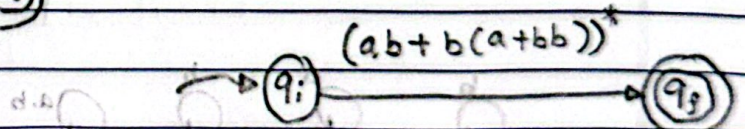
Step 3

Remove C



Step 4

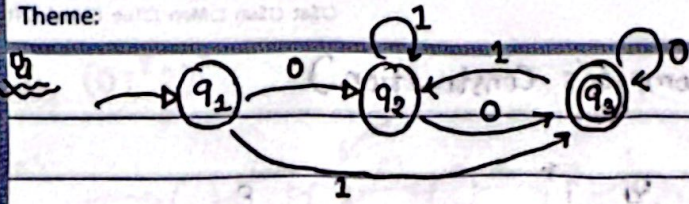
Remove A



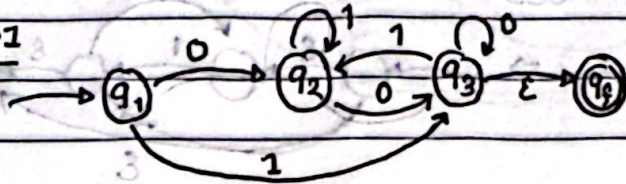
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⇒ Step 1



Step 2

Remove q_2

$q_1 \rightarrow q_3$

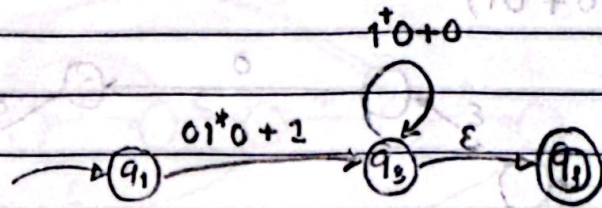
Self loop (q_3)

1. 01^*0

1. $11^*0 = 1^*0$

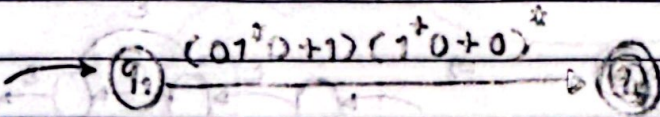
2. 1

2. 0



Step 3

Remove q_3



1. ...
2. ...
3. ...

papers King

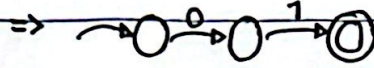
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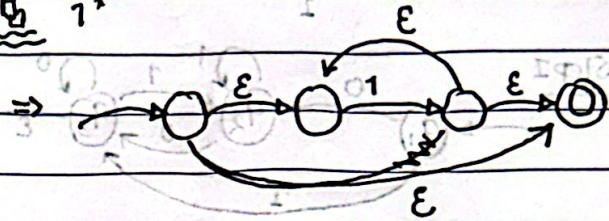
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Regular Expression to NFA (Thompson's Construction)

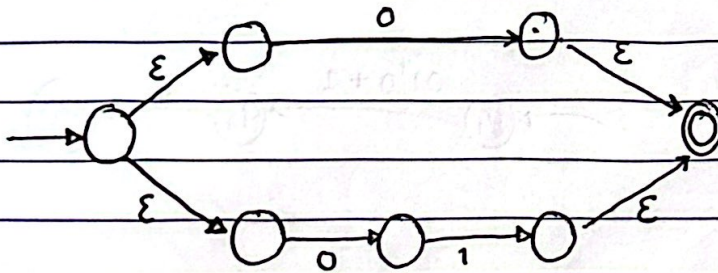
Q1



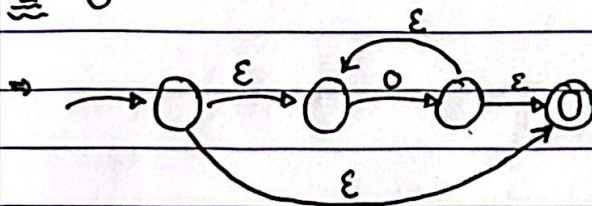
Q2 1^*



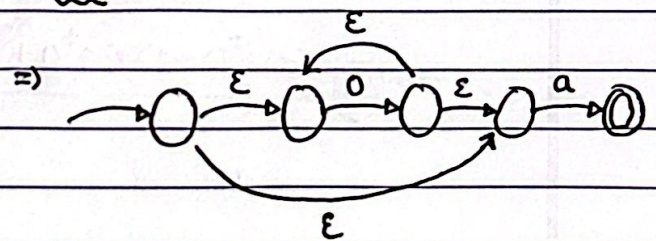
Q3 $(0 + 01)$



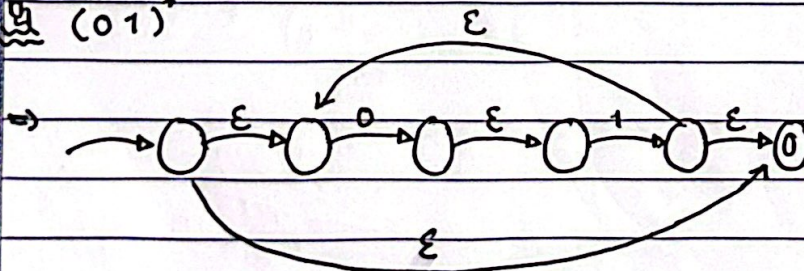
Q4 0^*



Q5 0^*a



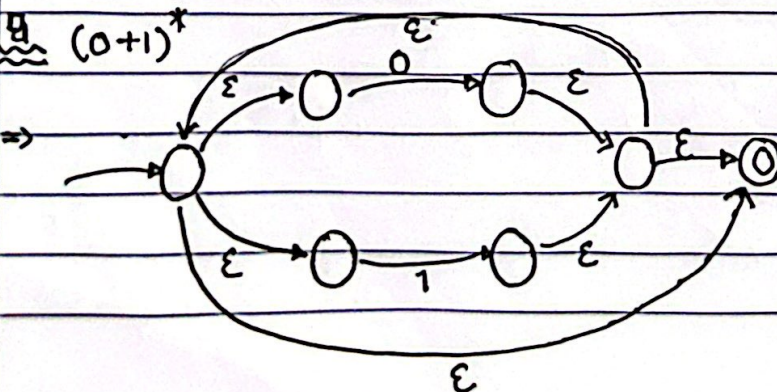
Q6 $(01)^*$



For Kleen Star (*)

1. Make the loop
2. Break the loop
3. Over the loop

Q7 $(0+1)^*$

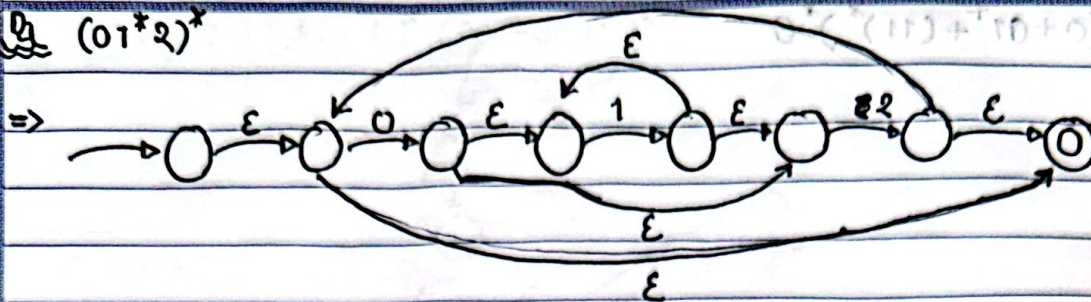


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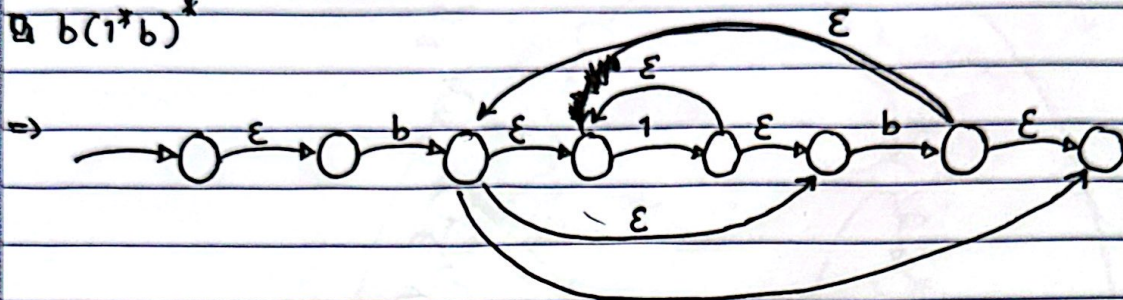
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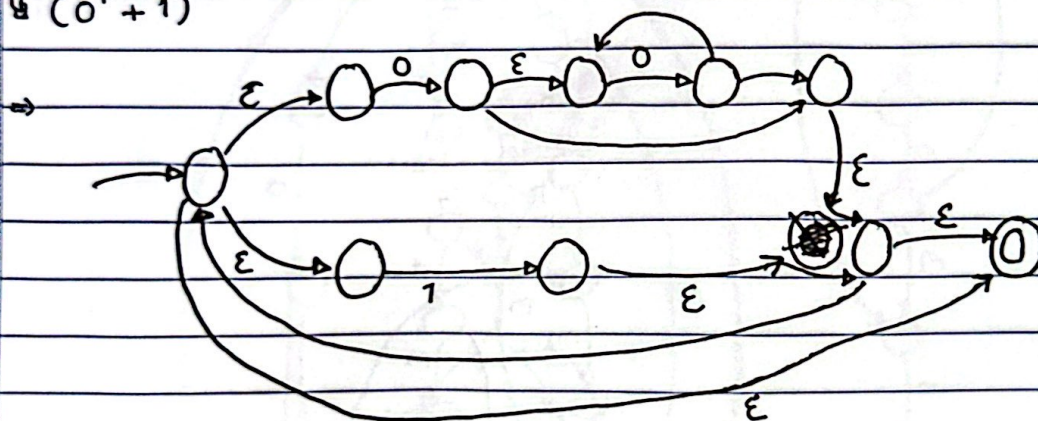
Q (01*2)*



Q b(1*b)



Q (0*+1)*



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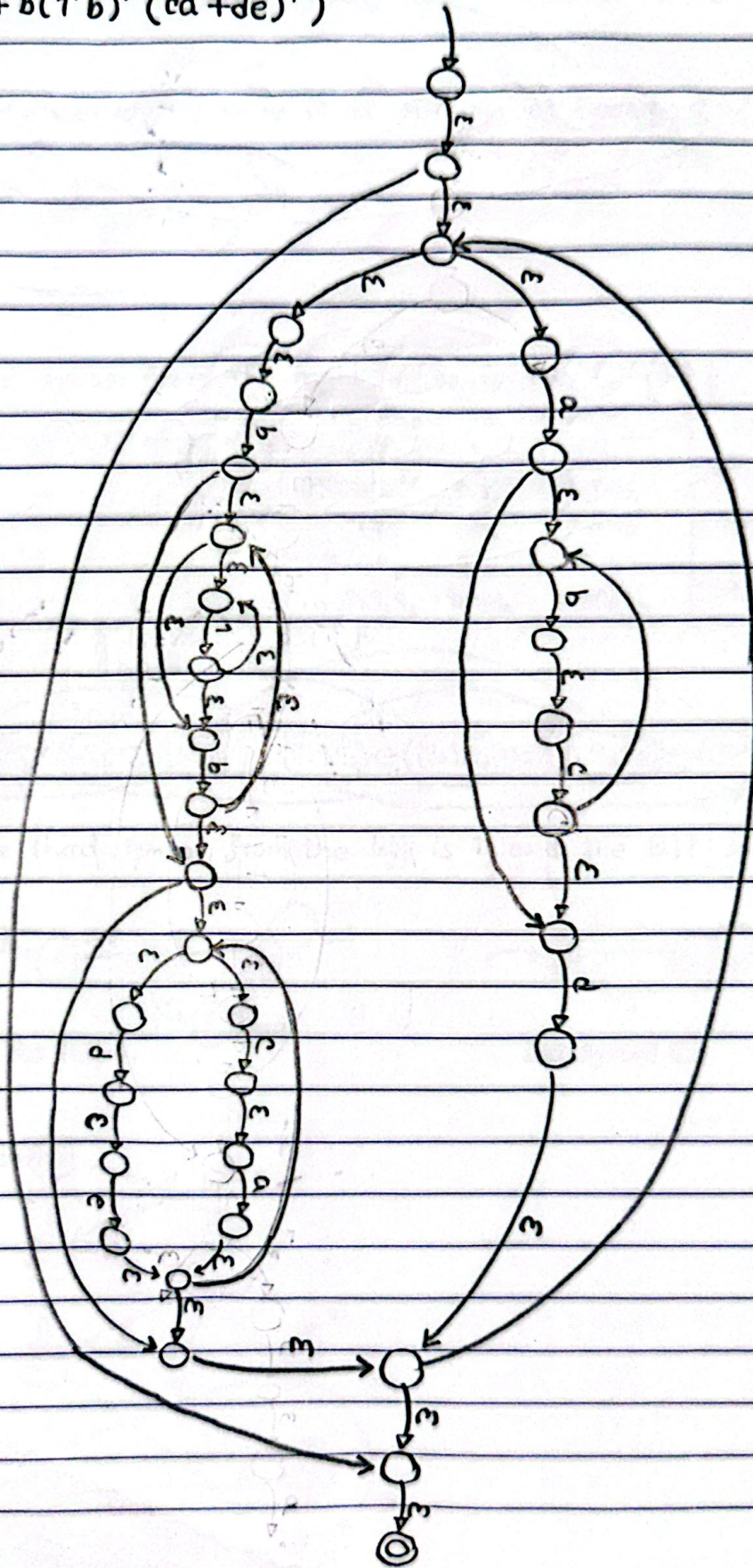
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$$(a(bc)^*d + b(1^*b)^*(ca + de)^*)^*$$

→



$$(a(bc)^*d + b(1^*b)^*(ca + de)^*)^*$$